Jon Arrizabalaga

Postdoctoral Researcher
Massachusetts Institute of Technology

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RESEARCH FOCUS

My research aims to enable autonomous systems that adapt to new tasks and conditions while providing explicit assurances about their safety and performance. To achieve this, I develop interpretable and high-performance algorithms that integrate control theory, numerical optimization, and artificial intelligence to model, plan, and control the motion of robotic and autonomous systems in complex, real-world environments.

PROFESSIONAL EXPERIENCE

ACADEMIC EXPERIENCE

Postdoctoral Researcher 2026 - present
Massachusetts Institute of Technology - AeroAstro *Cambridge, USA*

Visiting Ph.D. / Postdoctoral Researcher 2024 - 2026
Carnegie Mellon University - Robotics Institute *Pittsburgh, USA*

INDUSTRY EXPERIENCE

Bosch Research 2020
M.Sc. Thesis - Robotics Researcher [Paper], [Video] *Renningen, Germany*

Porsche AG 2019
Research Intern *Weissach, Germany*

EDUCATION

Ph.D. in Aerospace (Robotics & Controls) 2020 - 2025
Technical University of Munich [Thesis], [Defense] *Munich, Germany*
Visiting Charles University (2023)

M.Sc. in Mechatronics Engineering 2018 - 2020
KTH Royal Institute of Technology [Thesis] *Stockholm, Sweden*

B.Sc. in Mechanical Engineering 2014 - 2018
University of Navarre *San Sebastian, Spain*
Visiting Shanghai Jiao Tong University (2017 - 2018)

HONORS, AWARDS, AND FELLOWSHIPS

Carnegie Bosch Postdoctoral Fellowship *(declined)* [Program] 2025
Two-year postdoctoral research fellowship awarded to six recipients worldwide.

Best RoboCup Paper Award *(finalist)* 2024
5 nominations out of ~4000 submissions at the conference IEEE/RSJ IROS [Paper], [Talk]

Best Ph.D. Lecturer Award 2022
Recognizes the Ph.D. lecturer who has received the highest evaluations from B.Sc. and M.Sc. students

International Connecting Talent Scholarship 2019
Provided financial support during M.Sc. studies

Outstanding End-of-Degree Award *(finalist)* 2018
Honors B.Sc. students who have excelled in both their academic and extracurricular endeavors

PUBLICATIONS

PUBLISHED AND ACCEPTED

[P1] **A Differentiable Interior-Point Method in Single Precision**
J. Arrizabalaga, K. Tracy, Z. Manchester
NeurIPS, 2026 [Paper] [Code] [Website]

[P2] **Implicit Primal-Dual Interior-Point Methods for Quadratic Programming**
J. Arrizabalaga, Z. Manchester
IEEE CDC, 2026 [Paper] [Code]

[P3] **ADMM-based Continuous Trajectory Optimization in Graphs of Convex Sets**
L. Pries, **J. Arrizabalaga**, Z. Manchester, M. Ryll
IEEE IROS, 2026 [Paper]

[P4] **TinySDP: Real Time Semidefinite Optimization for Certifiable and Agile Edge Robotics**
I. Mahajan, **J. Arrizabalaga**, A. Grillo, F. Vega, J. Anderson, Z. Manchester, B. Plancher
RSS, 2026 [Paper] [Code] [Website]

[P5] **The Trajectory Bundle Method: Unifying Sequential-Convex Programming and Sampling-Based Trajectory Optimization**
K. Tracy, J. Z. Zhang, **J. Arrizabalaga**, S. Schaal, Y. Tassa, T. Erez, Z. Manchester
IEEE ICRA, 2026 [Paper]

[P6] **Conflict-Based Search as a Protocol: A Multi-Agent Motion Planning Protocol for Heterogeneous Agents, Solvers, and Independent Tasks**
R. Veerapaneni, A. Tang, H. He, S. Zhao, V. Shah, Y. Cen, Z. Ji, G. Olin, **J. Arrizabalaga**, Y. Shaoul, J. Li, M. Likhachev
IEEE ICRA, 2026 [Paper] [Website]

[P7] **Trajectory Planning Using Safe Ellipsoidal Corridors as Projections of Orthogonal Trust Regions**
A. Jaitly, **J. Arrizabalaga**, G. Li
IEEE ICRA, 2026 [Paper]

[P8] **A Universal Formulation for Path-Parametric Planning and Control: A Unified Framework, from Path following to Contouring Control and Progress Maximization**
J. Arrizabalaga, Z. Šír, Z. Manchester, M. Ryll
IEEE CSM, 2026 [Paper] [Code] [Website] [Talk]

[P9] **DQ-NMPC: Dual-Quaternion NMPC for Quadrotor Flight**
L.F. Recalde, D. Agrawal, **J. Arrizabalaga**, G. Li
RA-L, 2025 [Paper] [Video] [Code] [Website]

[P10] **Convex Maneuver Planning for Spacecraft Collision Avoidance**
F. Vega, **J. Arrizabalaga**, R. Watson, Z. Manchester
IEEE iSpaRo, 2025 [Paper] [Code]

[P11] **PHODCOS: Pythagorean Hodograph-based Differentiable Coordinate System**
J. Arrizabalaga, F. Vega, Z. Šír, Z. Manchester, M. Ryll
IEEE Aero, 2025 [Paper] [Code] [Talk]

[P12] **Differentiable Collision-Free Parametric Corridors**
J. Arrizabalaga, Z. Manchester, M. Ryll
IEEE IROS, 2024 Best Paper Finalist. [Paper] [Code] [Video] [Talk]

PUBLICATIONS (CONTINUED)

PUBLISHED AND ACCEPTED (CONTINUED)

[P13] **Geometric Slosh-Free Tracking for Robotic Manipulators**
J. Arrizabalaga, L. Pries, R. Laha, R. Li, S. Haddadin, M. Ryll
IEEE ICRA, 2024 [Paper] [Code] [Video] [Talk]

[P14] **Learning for CasADi: Data-Driven Models in Numerical Optimization**
T. Salzmann, **J. Arrizabalaga**, J. Andersson, M. Pavone, M. Ryll
L4DC, 2024 [Paper] [Code] [Talk]

[P15] **Pose-Following with Dual Quaternions**
J. Arrizabalaga, M. Ryll
IEEE CDC, 2023 [Paper] [Code] [Talk]

[P16] **SCTOMP: Spatially Constrained Time-Optimal Motion Planning**
J. Arrizabalaga, M. Ryll
IEEE IROS, 2023 [Paper] [Video]

[P17] **Real-Time Neural MPC: Deep Learning Model Predictive Control for Quadrotors and Agile Robotic Platforms**
T. Salzmann, E. Kaufmann, **J. Arrizabalaga**, M. Pavone, D. Scaramuzza, M. Ryll
RA-L, 2023 [Paper] [Code]

[P18] **Spatial Motion Planning with Pythagorean Hodograph curves**
J. Arrizabalaga, M. Ryll
IEEE CDC, 2022 [Paper] [Video] [Talk]

[P19] **Towards Time-Optimal Tunnel-Following for Quadrotors**
J. Arrizabalaga, M. Ryll
IEEE ICRA, 2022 [Paper] [Video]

[P20] **A caster-wheel-aware MPC-based motion planner for mobile robotics**
J. Arrizabalaga, N. van Duijkeren, M. Ryll, R. Lange
IEEE ICAR, 2021 [Paper] [Thesis] [Video]

MANUSCRIPTS UNDER REVIEW

[U1] **Graphs of Trajectory Bundles for Nonlinear Motion Planning**
J. Verhagen, **J. Arrizabalaga**, J. Tumova, Z. Manchester
Under Review, 2026 [Paper]

[U2] **ElastiQP: An Always-Feasible QP Solver for Constrained Robot Control**
D. Morton, **J. Arrizabalaga**, Z. Manchester, M. Pavone
Under Review, 2026 [Paper] [Code]

[U3] **Fast and Robust Temporal Logic Planning via ADMM-based Trajectory Optimization**
L. Pries, J. Verhagen, **J. Arrizabalaga**, J. Tumova, M. Ryll, Z. Manchester
Under Review, 2026 [Paper]

[U4] **End-to-end QP-based policies: A unified perspective on robust control and robot learning**
F. Vega, P. S. Balaji, C. Dunaway, J. Koszut, **J. Arrizabalaga**, Z. Manchester
Under Review, 2026 [Paper] [Website]

INVITED TALKS

Control Group Seminar, *University of Oxford* 2026

Symposium on Society for Industrial and Applied Mathematics, *UMass Dartmouth* 2026

Northeast Systems and Control Workshop, *Columbia University* 2025

Talking Robotics, Virtual Seminar [Talk] 2025

GRASP Laboratory, Vijay Kumar Lab, *University of Pennsylvania* 2024

German Aerospace Center (DLR), Institute of Flight Systems 2023

Robotics and Perception Group, *University of Zurich* 2022

TEACHING EXPERIENCE

Dynamics 2026
Co-Instructor, Massachusetts Institute of Technology
Level: B.Sc., *Students*: 50

Engineering Mechanics I 2021 - 2024
Lecturer, Technical University of Munich
Level: B.Sc., *Students*: 200 ~ 450

Mechanics for Aerospace 2021 - 2023
Lecturer, Technical University of Munich (Asia)
Level: M.Sc., *Students*: 20 ~ 25

Introduction to ROS (Robot Operating System) 2021 - 2022
Lecturer, Technical University of Munich
Level: M.Sc., *Students*: 50

Dynamics Motion and Control 2019 - 2020
Teaching Assistant, KTH Royal Institute of Technology
Level: M.Sc., *Students*: 50

Robust Mechatronics 2019 - 2020
Teaching Assistant, KTH Royal Institute of Technology
Level: M.Sc., *Students*: 50

MENTORING

STUDENT MENTORING

Priyanka Supraja Balaji, Research Fellow · *now at California Institute of Technology* 2026

Viraj Shah, Research Project · *now at Carnegie Mellon University* 2025

Eden Cen, Research Project · *now at Carnegie Mellon University* 2025

Akshay Jaitly, Research Project · *now at Worcester Polytechnic Institute* 2025

Luis Fernando Recalde, Research Project · *now at Worcester Polytechnic Institute* 2024

Giuseppe Soldati, Semester Thesis · *now at German Aerospace Center (DLR)* 2023

Harun Tongay Tamtürk, Semester Thesis & M.Sc. Thesis · *now at Airbus Defence and Space* 2023

Bruno Sorban, Research Project · *now at Boston Consulting Group* 2023

Tommaso Faraci, Research Project · *now at Universita di Trento* 2022

TEAM MENTORING

ASCENT - Rocket Hopper, Guidance and Control Advisor [Website] 2022 - 2024
Advised the development of optimization-based trajectory planning and model predictive flight control for a rocket-hopper platform

SERVICE

WORKSHOP ORGANIZATION

Designing Well-Behaved Optimization Problems for Learning and Control in Robotics 2026
Symposium on Society for Industrial and Applied Mathematics [Website]
Organizers: G. Bravo, *J. Arrizabalaga*, F. Dümbgen, and S. Heim

Workshop on the Path Towards Generalizable Contact-Rich Robotics: Control and Representation 2026
IEEE International Conference on Robotics and Automation (ICRA) [Website]
Organizers: A. Bishop, G. Sarmiento, J. Zhang, *J. Arrizabalaga*, M. Posa, P. Culbertson, S. Le Cleach, X. Cheng, Z. Manchester

PEER REVIEW

Journals: IEEE Transactions on Robotics (T-RO); IEEE Transactions on Robot Learning (T-RL); IEEE Robotics and Automation Letters (RA-L); IEEE Transactions on Systems, Man, and Cybernetics: Systems (T-SMC); The International Journal of Robotics Research (IJRR).

Conferences: Robotics Science and Systems (RSS); IEEE International Conference on Robotics and Automation (ICRA); IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS); IEEE Conference on Decision and Control (CDC); IEEE International Conference on Humanoid Robots (Humanoids); IEEE International Conference on Space Robotics (iSpaRo).